

Question-Type Index

All 180 Questions – Subject, Difficulty, Marks & Problem Type

Legend: ✓ Correct (+4) ✗ Incorrect (−1) – Unattempted (0)

Physics (Q1–45)

Q#	Subject	Difficulty	Marks	Result	Problem Type
1	Physics	EASY	+4	✓	Calculus – rate of change (differentiation)
2	Physics	EASY	-1	✗	Clock-angle geometry
3	Physics	EASY	0	–	Trigonometry – inclined-plane height problem
4	Physics	EASY	+4	✓	Quadratic function analysis (vertex, roots, slope)
5	Physics	MEDIUM	-1	✗	3D vector displacement & resultant
6	Physics	EASY	+4	✓	River–boat relative velocity (crossing time)
7	Physics	EASY	+4	✓	Vector addition of coplanar forces
8	Physics	EASY	+4	✓	Relative velocity – component along a direction
9	Physics	MEDIUM	0	–	Vector projection / component problem
10	Physics	MEDIUM	+4	✓	3D geometry – tetrahedral bond-angle vectors
11	Physics	MEDIUM	-1	✗	Relative velocity resolved along shoreline
12	Physics	MEDIUM	+4	✓	Curved motion – distance travelled
13	Physics	EASY	+4	✓	Vector force equilibrium (missing force)
14	Physics	MEDIUM	+4	✓	Vector dot/cross product – list matching
15	Physics	HARD	0	–	Kinematics – acceleration from implicit position function
16	Physics	EASY	+4	✓	Kinematics – velocity from a–t graph
17	Physics	MEDIUM	+4	✓	Kinematics – free-fall window-passage problem
18	Physics	MEDIUM	0	–	Relative motion – projectile from accelerating balloon
19	Physics	HARD	+4	✓	Vertical projectile motion (statement-based)
20	Physics	MEDIUM	+4	✓	Projectile + pursuit problem (ball & dog)
21	Physics	MEDIUM	+4	✓	Projectile motion – comparing two trajectories
22	Physics	HARD	-1	✗	2D motion with horizontal wind acceleration
23	Physics	MEDIUM	-1	✗	Relative velocity – aircraft & wind drift
24	Physics	HARD	0	–	Monkey-hunter projectile problem
25	Physics	MEDIUM	0	–	Vector condition for collision of moving particles
26	Physics	HARD	+4	✓	Average velocity vs average speed (x–y graph)
27	Physics	HARD	+4	✓	Average velocity vs average speed (x–y graph)
28	Physics	EASY	0	–	Projectile motion – time above given height
29	Physics	EASY	+4	✓	Assertion–Reason – projectile motion physics
30	Physics	MEDIUM	0	–	Vector algebra – angle between vectors
31	Physics	EASY	0	–	Coplanar vectors – scalar triple product
32	Physics	MEDIUM	-1	✗	Unit vector perpendicular to a plane (cross product)
33	Physics	EASY	+4	✓	Average acceleration – direction change at constant speed
34	Physics	MEDIUM	0	–	1D kinematics – implicit position-time relation
35	Physics	MEDIUM	+4	✓	Projectile motion – max height from velocity ratio
36	Physics	MEDIUM	+4	✓	Projectile motion – vertical velocity at half max height
37	Physics	EASY	0	–	Motion in a velocity field – path equation (diff. eq.)

Q#	Subject	Difficulty	Marks	Result	Problem Type
38	Physics	EASY	+4	✓	Projectile motion – angle of projection from graph
39	Physics	MEDIUM	-1	✗	Integration – indefinite integral of trig function
40	Physics	EASY	+4	✓	Vector representation of a force along a given line
41	Physics	MEDIUM	-1	✗	1D kinematics – distance travelled (turning point)
42	Physics	HARD	0	–	Curved motion – average acceleration from velocity vectors
43	Physics	MEDIUM	+4	✓	Free fall – regularly released water drops
44	Physics	MEDIUM	+4	✓	Relative motion – dropped particle from balloon (match columns)
45	Physics	HARD	0	–	Kinematics – car race relative speed/time problem

Chemistry (Q46–90)

Q#	Subject	Difficulty	Marks	Result	Problem Type
46	Chemistry	EASY	+4	✓	Mole concept – atom counting in a compound
47	Chemistry	EASY	+4	✓	Mole concept – gas volume at STP
48	Chemistry	EASY	+4	✓	Mole concept – % composition & molecular mass
49	Chemistry	EASY	+4	✓	Mole concept – moles remaining after removal
50	Chemistry	EASY	+4	✓	Mole concept – vapour density from STP volume
51	Chemistry	EASY	+4	✓	Stoichiometry – combustion of methane
52	Chemistry	EASY	+4	✓	Mole concept – mass of a single atom
53	Chemistry	EASY	+4	✓	Assertion–Reason – atomic mass unit (amu)
54	Chemistry	HARD	+4	✓	Molarity / solution preparation calculation
55	Chemistry	MEDIUM	+4	✓	Combustion analysis – molecular formula of hydrocarbon
56	Chemistry	HARD	+4	✓	Stoichiometry – % purity from thermal decomposition
57	Chemistry	MEDIUM	+4	✓	Mole concept – atom ratio from atomic weights
58	Chemistry	HARD	+4	✓	Empirical formula determination
59	Chemistry	HARD	-1	✗	Stoichiometry – list matching (balanced equation)
60	Chemistry	HARD	+4	✓	Molarity from density & mass percentage
61	Chemistry	EASY	+4	✓	Atomic structure – electrons/protons in an ion
62	Chemistry	MEDIUM	+4	✓	Photon energy – frequency–wavelength calculation
63	Chemistry	EASY	+4	✓	Isotopic abundance & average atomic weight
64	Chemistry	EASY	-1	✗	Nuclear physics – nuclear diameter from mass number
65	Chemistry	MEDIUM	+4	✓	Bohr model – orbital energy of hydrogenic ion
66	Chemistry	EASY	+4	✓	Bohr model – scope/limitations (conceptual)
67	Chemistry	EASY	0	–	Bohr model – time-period ratio (H vs He■)
68	Chemistry	HARD	0	–	Photon energy conservation (absorption/emission)
69	Chemistry	MEDIUM	+4	✓	Atomic structure – conceptual statement evaluation
70	Chemistry	MEDIUM	-1	✗	Atomic structure – identify incorrect statement
71	Chemistry	MEDIUM	+4	✓	Isoelectronic species – ordering by proton number
72	Chemistry	HARD	-1	✗	Electronic configuration – statement evaluation
73	Chemistry	HARD	0	–	Heisenberg uncertainty principle (position–velocity)
74	Chemistry	HARD	+4	✓	Hydrogen spectrum – Balmer series formula
75	Chemistry	MEDIUM	0	–	Heisenberg uncertainty principle – numerical
76	Chemistry	HARD	+4	✓	Periodic table – element ID from electronic configuration
77	Chemistry	MEDIUM	+4	✓	Bond polarity – diagram interpretation
78	Chemistry	MEDIUM	0	–	Ionisation energy – stepwise IE calculation
79	Chemistry	MEDIUM	-1	✗	Periodic trends – IP vs atomic number graph
80	Chemistry	EASY	+4	✓	Periodic trends – EA, EN, IP comparison from configs
81	Chemistry	MEDIUM	-1	✗	Thermodynamics – ionisation/electron-affinity energy diagram

Q#	Subject	Difficulty	Marks	Result	Problem Type
82	Chemistry	MEDIUM	+4	✓	Periodic table history – statement evaluation
83	Chemistry	HARD	+4	✓	Acidic strength of oxides (periodic trend)
84	Chemistry	HARD	+4	✓	Periodic trends – 2nd ionisation energy ordering
85	Chemistry	MEDIUM	+4	✓	IUPAC nomenclature matching (superheavy elements)
86	Chemistry	MEDIUM	+4	✓	Periodic properties – identify incorrect statement
87	Chemistry	HARD	+4	✓	Periodic trends – Slater's rule & IP jump
88	Chemistry	MEDIUM	+4	✓	Periodic trends – identify incorrect order
89	Chemistry	MEDIUM	+4	✓	Electronic configuration – identify element/block
90	Chemistry	MEDIUM	+4	✓	Periodic trends – electronegativity ordering

Biology – Section A (Q91–120)

Q#	Subject	Difficulty	Marks	Result	Problem Type
91	Biology (A)	EASY	-1	✗	Diagram-based classification identification
92	Biology (A)	EASY	+4	✓	Bacterial cell structure (attachment structures)
93	Biology (A)	MEDIUM	-1	✗	Taxonomic hierarchy – identify unrelated taxon
94	Biology (A)	EASY	+4	✓	Taxonomic hierarchy – plant/animal rank analogy
95	Biology (A)	MEDIUM	+4	✓	Taxonomic hierarchy – shared-character trend
96	Biology (A)	HARD	+4	✓	Kingdom Monera – statement evaluation
97	Biology (A)	MEDIUM	+4	✓	Taxonomic hierarchy – column matching (wheat)
98	Biology (A)	MEDIUM	+4	✓	Classification terms – column matching
99	Biology (A)	EASY	+4	✓	Endomembrane system – odd-one-out function
100	Biology (A)	EASY	+4	✓	Cell organelles – statement evaluation (microbodies)
101	Biology (A)	EASY	+4	✓	Mitochondria structure – statement evaluation
102	Biology (A)	EASY	+4	✓	Cell biology – Golgi–ER association (reasoning)
103	Biology (A)	MEDIUM	-1	✗	Chromatin/chromosome structure – statement count
104	Biology (A)	MEDIUM	-1	✗	Centriole vs cilia/flagella – structural comparison
105	Biology (A)	HARD	+4	✓	Meiosis – Prophase-I statement evaluation
106	Biology (A)	HARD	+4	✓	Cell theory history – three-column matching
107	Biology (A)	MEDIUM	+4	✓	Assertion–Reason – semi-autonomous organelles
108	Biology (A)	EASY	+4	✓	Cell organelles – membrane-less structure count
109	Biology (A)	EASY	+4	✓	Cell organelles – correct matching
110	Biology (A)	EASY	+4	✓	Assertion–Reason – cytoplasm as site of activity
111	Biology (A)	EASY	+4	✓	Chromosome morphology – identify types from diagram
112	Biology (A)	MEDIUM	+4	✓	Cell structure – bacteria vs plant cell (common feature)
113	Biology (A)	MEDIUM	-1	✗	Cell division – flow-chart interpretation (ploidy)
114	Biology (A)	MEDIUM	+4	✓	Meiosis – identify stage (Anaphase I)
115	Biology (A)	MEDIUM	-1	✗	Meiosis – sequence of events in Prophase I
116	Biology (A)	MEDIUM	+4	✓	Assertion–Reason – mitosis & growth
117	Biology (A)	MEDIUM	+4	✓	Meiosis – sequence of Prophase-I sub-stages
118	Biology (A)	MEDIUM	-1	✗	Cell division – identify stage from diagram
119	Biology (A)	HARD	+4	✓	Meiosis – DNA content after Meiosis-I
120	Biology (A)	MEDIUM	-1	✗	Meiosis – chromatid count in Prophase I

Biology – Section B (Q121–180)

Q#	Subject	Difficulty	Marks	Result	Problem Type
121	Biology (B)	HARD	+4	✓	Assertion–Reason – taxonomic family/genus classification
122	Biology (B)	MEDIUM	+4	✓	Taxonomic hierarchy – statement evaluation
123	Biology (B)	HARD	+4	✓	Taxonomic hierarchy – higher vs lower taxa
124	Biology (B)	EASY	+4	✓	Growth – identify incorrect statement
125	Biology (B)	EASY	+4	✓	Taxonomic rank matching (division/class/order/family)
126	Biology (B)	EASY	+4	✓	Reproduction as a life characteristic – statement evaluation
127	Biology (B)	EASY	+4	✓	Classical vs modern taxonomy – statement comparison
128	Biology (B)	EASY	+4	✓	Classification terms – column matching
129	Biology (B)	MEDIUM	+4	✓	Kingdom Protista – identify incorrect statement
130	Biology (B)	HARD	+4	✓	Statement evaluation – Mycoplasma & bacterial structure
131	Biology (B)	EASY	+4	✓	Bacterial cell wall/protective layer identification
132	Biology (B)	HARD	-1	✗	Three-domain classification system – statement evaluation
133	Biology (B)	EASY	+4	✓	Five-kingdom classification – criteria identification
134	Biology (B)	EASY	+4	✓	Mycoplasma – characteristics statement evaluation
135	Biology (B)	EASY	+4	✓	Phylogenetic classification – basis identification
136	Biology (B)	EASY	+4	✓	Mollusca – genus/animal matching
137	Biology (B)	EASY	+4	✓	Platyhelminthes – identify incorrect statement
138	Biology (B)	EASY	+4	✓	Animal body plan – triploblastic acoelomate phylum
139	Biology (B)	HARD	+4	✓	Amphibia vs Reptilia – shared features
140	Biology (B)	EASY	+4	✓	Genus/character/phylum matching (odd one out)
141	Biology (B)	EASY	+4	✓	Tissue-level organisation – animal identification
142	Biology (B)	EASY	+4	✓	Animal classification – matching (incorrect pair)
143	Biology (B)	MEDIUM	+4	✓	Pisces – operculum/air-bladder absence
144	Biology (B)	EASY	+4	✓	Cyclostomata – characteristic identification
145	Biology (B)	MEDIUM	+4	✓	Viviparous animals – identification set
146	Biology (B)	EASY	+4	✓	Coelenterata – exoskeleton identification
147	Biology (B)	EASY	+4	✓	Amphibia – characteristics identification
148	Biology (B)	EASY	+4	✓	Porifera – spongocoel lining cell type
149	Biology (B)	EASY	0	–	Closed circulatory system – animal identification
150	Biology (B)	EASY	+4	✓	Aquatic mammals – correct combination
151	Biology (B)	EASY	0	–	Chordates lacking notochord/vertebral column (adult)
152	Biology (B)	MEDIUM	0	–	Assertion–Reason – Arthropoda as largest phylum
153	Biology (B)	MEDIUM	0	–	Animal classification – same-phylum grouping
154	Biology (B)	MEDIUM	0	–	Homeothermy – identify non-homeotherm set
155	Biology (B)	MEDIUM	0	–	Animal pairs – matching shared feature
156	Biology (B)	MEDIUM	0	–	Monoecious animals – group identification
157	Biology (B)	HARD	0	–	Poikilothermic animals – count from list

Q#	Subject	Difficulty	Marks	Result	Problem Type
158	Biology (B)	MEDIUM	0	—	Nereis vs leech – comparative features
159	Biology (B)	MEDIUM	-1	✗	Coelom types – statement evaluation
160	Biology (B)	EASY	0	—	Internal fertilization – animal identification
161	Biology (B)	EASY	+4	✓	Parasitology – identify incorrect statement
162	Biology (B)	EASY	+4	✓	Animal pairing – correct match
163	Biology (B)	EASY	0	—	Filarial worm – characteristic identification
164	Biology (B)	EASY	+4	✓	Mollusca – mantle/foot/shell identification
165	Biology (B)	EASY	-1	✗	Echinodermata – sea lily classification
166	Biology (B)	EASY	-1	✗	Marine vs non-marine animal groups
167	Biology (B)	EASY	+4	✓	Jaws presence – animal group identification
168	Biology (B)	EASY	0	—	Chordata subphyla – notochord statement evaluation
169	Biology (B)	EASY	+4	✓	Diagram-based animal & phylum identification
170	Biology (B)	EASY	+4	✓	Notochord presence – group identification
171	Biology (B)	EASY	+4	✓	Marine fish identification
172	Biology (B)	EASY	-1	✗	Diagram-based class identification (fish groups)
173	Biology (B)	EASY	+4	✓	Taxonomic classification of Rana tigrina (frog)
174	Biology (B)	EASY	+4	✓	Diagram-based animal identification
175	Biology (B)	EASY	+4	✓	Reptilia – characteristic identification
176	Biology (B)	EASY	+4	✓	Bird scales – location identification
177	Biology (B)	EASY	0	—	Exclusively marine animal sets
178	Biology (B)	EASY	-1	✗	Porifera – osculum/ostia (mouth-equivalent) identification
179	Biology (B)	EASY	+4	✓	Taxon with both freshwater & marine species
180	Biology (B)	EASY	+4	✓	Pisces – bony fish count from list